

Ion Save - Software Architecture Design

System Overview

Ion Save is a mobile-first application designed to digitize Ikimina (traditional Rwandan community savings groups) with support for 1,500+ users in the first 12 months, scalable to tens of thousands.

Architecture Components

1. Mobile Application Layer

Technology Stack:

- **Frontend:** React Native or Flutter for cross-platform (iOS/Android)
- **State Management:** Redux/MobX or Riverpod (Flutter)
- **Offline Storage:** SQLite with sync capabilities
- **Authentication:** Biometric + PIN authentication

Key Features:

- Multilingual support (Kinyarwanda, French, English)
- Offline-first design for rural connectivity
- Group management interface
- Contribution tracking
- Loan management
- Financial reporting

2. Backend Services (Microservices Architecture)

Core Services:

User Management Service

- User authentication & authorization
- Profile management
- Role-based access control (Admin, Member, Treasurer)
- JWT token management

Group Management Service

- Ikimina group creation & management
- Member enrollment & removal
- Group settings & rules
- Meeting scheduling

Financial Transaction Service

- Contribution tracking
- Loan processing & approval workflows
- Interest calculations
- Transaction history
- Payment gateway integration

Notification Service

- Push notifications
- SMS alerts (for non-smartphone users)
- Email notifications
- Meeting reminders

Reporting & Analytics Service

- Financial reports generation
- Group performance analytics
- Individual member summaries
- Export functionality (PDF, CSV)

3. Database Architecture

Primary Database: PostgreSQL

- **Users Table:** User profiles, authentication data
- **Groups Table:** Ikimina group information
- **Members Table:** Group membership relationships
- **Transactions Table:** All financial transactions
- **Loans Table:** Loan records and repayment schedules
- **Meetings Table:** Meeting records and attendance

Cache Layer: Redis

- Session management
- Frequently accessed group data
- Real-time transaction processing

Document Storage: AWS S3 or Google Cloud Storage

- User profile photos
- Group documents
- Transaction receipts
- Report exports

4. API Gateway & Load Balancing

API Gateway (Kong or AWS API Gateway)

- Request routing
- Rate limiting
- Authentication validation
- API versioning
- Request/response transformation

Load Balancer

- Distribute traffic across service instances
- Health checks
- Auto-scaling triggers

5. Payment Integration Layer

Mobile Money Integration

- MTN MoMo API
- Airtel Money API
- Tigo Cash API (if applicable)

Banking Integration

- Local bank APIs (Bank of Kigali, Equity Bank Rwanda)
- SWIFT integration for international transfers

- PCI DSS compliance

6. Security & Compliance

Security Measures:

- End-to-end encryption for sensitive data
- HTTPS/TLS 1.3 for all communications
- Multi-factor authentication
- Regular security audits
- GDPR compliance for data protection

Compliance:

- National Bank of Rwanda regulations
- Anti-money laundering (AML) compliance
- Know Your Customer (KYC) requirements

7. Infrastructure & Deployment

Cloud Platform: AWS or Google Cloud Platform

- **Compute:** Auto-scaling EC2 instances or Google Compute Engine
- **Container Orchestration:** Kubernetes or Docker Swarm
- **CI/CD:** GitHub Actions or GitLab CI
- **Monitoring:** CloudWatch, Datadog, or New Relic

Environments:

- Development
- Staging
- Production
- DR (Disaster Recovery)

8. Offline Functionality

Offline-First Architecture:

- Local SQLite database with sync capabilities
- Conflict resolution mechanisms

- Background sync when connectivity restored
- Critical operations queued for retry

9. Scalability Considerations

Horizontal Scaling:

- Microservices can be scaled independently
- Database read replicas
- CDN for static assets
- Caching strategies

Performance Optimization:

- Database indexing
- Query optimization
- API response caching
- Image compression and lazy loading

10. Monitoring & Analytics

Application Monitoring:

- Real-time performance metrics
- Error tracking and alerting
- User behavior analytics
- System health dashboards

Business Intelligence:

- Financial transaction trends
- Group performance metrics
- User engagement analytics
- Regional usage patterns

Technology Stack Summary

Component	Technology Choice	Justification
Mobile App	React Native/Flutter	Cross-platform, faster development
Backend	Node.js/Express or Django	Rapid development, good ecosystem
Database	PostgreSQL	ACID compliance, JSON support
Cache	Redis	High performance, pub/sub support
Message Queue	RabbitMQ or Apache Kafka	Reliable message processing
File Storage	AWS S3/Google Cloud Storage	Scalable, cost-effective
Payment Gateway	Local mobile money APIs	Local market integration

Development Phases

Phase 1: Core MVP (Months 1-3)

- User authentication & basic profiles
- Group creation & management
- Basic contribution tracking
- Simple reporting

Phase 2: Financial Features (Months 4-6)

- Loan management system
- Mobile money integration
- Advanced reporting
- Meeting management

Phase 3: Scale & Optimize (Months 7-12)

- Performance optimization
- Advanced analytics
- Offline synchronization
- Additional payment methods

Security & Compliance Framework

Data Protection:

- Encryption at rest and in transit
- Regular security audits

- Compliance with local financial regulations
- User consent management

Financial Compliance:

- Transaction audit trails
- KYC verification processes
- AML monitoring
- Regulatory reporting capabilities

This architecture provides a solid foundation for Ion Save to start with 1,500+ users and scale to support tens of thousands while maintaining security, performance, and reliability standards required for financial applications.